Brief Introduction to Topological phases.

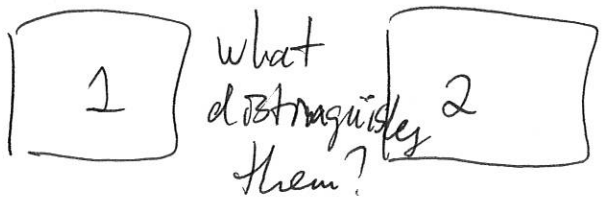
Symmetry arguments play key role in understanding different states of matter.

- Crystal order is distinguished from liquid (or gas) by spontaneous breaking of continuous translation symmetry. In crystal, only translations by multiple of lattice vectors transform the crystal to itself, while in liquid translation by any amount represents physical symmetry operation.

- Magnetic order: spontaneous breaking of continuous spin symmetry — spins pick up particular direction out of 4π available.

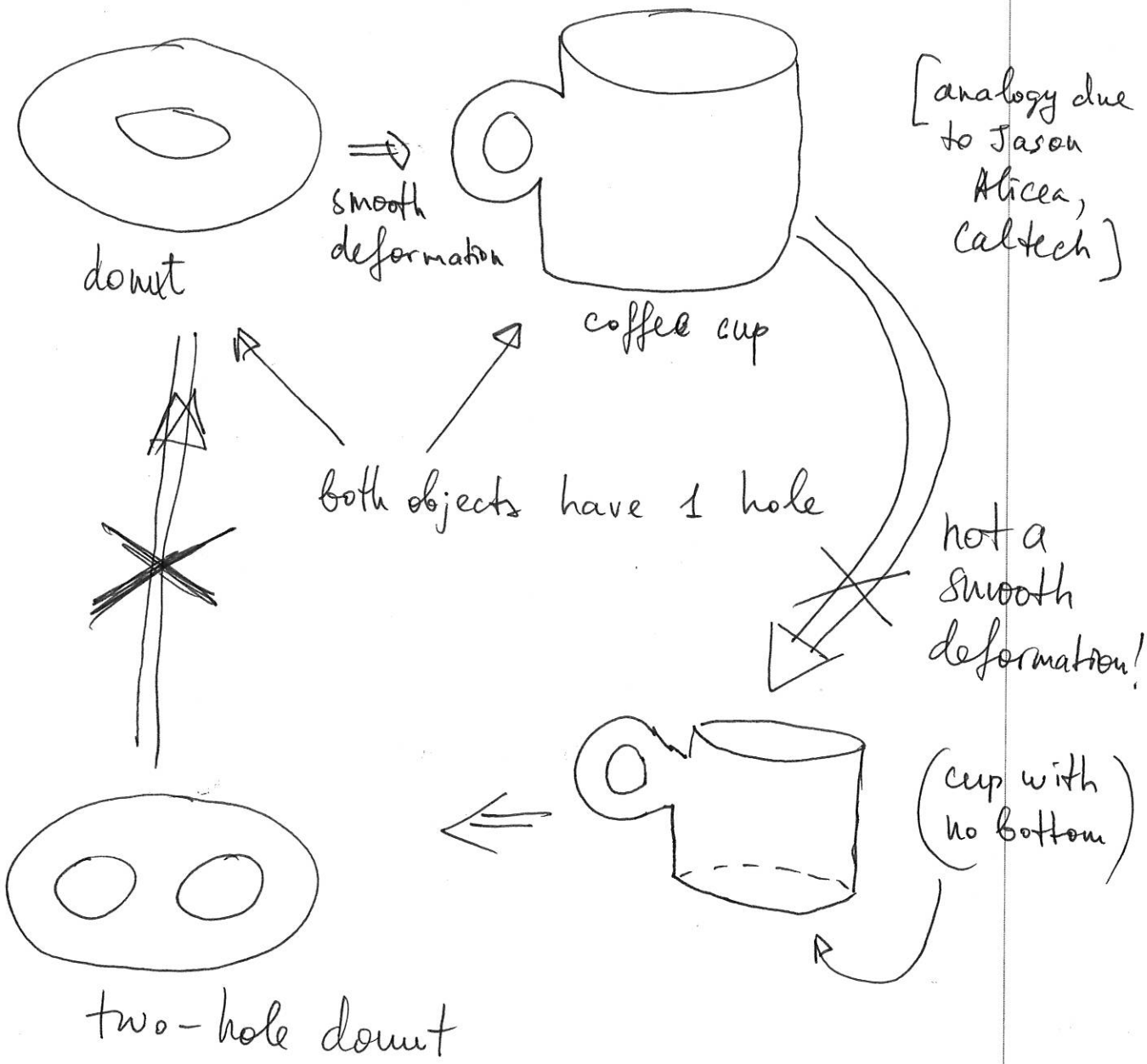
But in 1980's and later, largely due to quantum Hall effect discovery, it was realized that one can have <u>different</u> ~~phase~~ and <u>distinct</u> phases for which the concept of <u>local order parameter</u> does not apply.

→ and seemingly having the same symmetry properties.

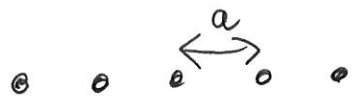


⇒ X. G. Wen: topological order.

Topology


donut
smooth deformation
coffee cup
[analogy due to Jason Alicea, Caltech]
both objects have 1 hole
not a smooth deformation!
(cup with no bottom)
two-hole donut

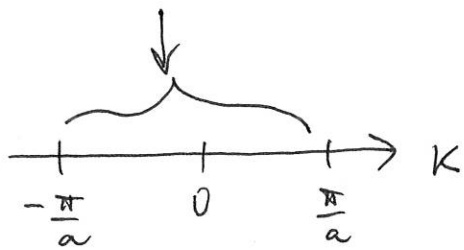
1d topological phase



1 electron per site
(half-filling condition)

$c^+_{x\sigma}$ creates electron with spin $S^z = \frac{\hbar}{2}\sigma$ at site $\underline{x}$.

$$c_{x\sigma} = \frac{1}{\sqrt{N}} \sum_{k \in BZ} e^{ikx} c_{k\sigma}$$



Hamiltonian in momentum space

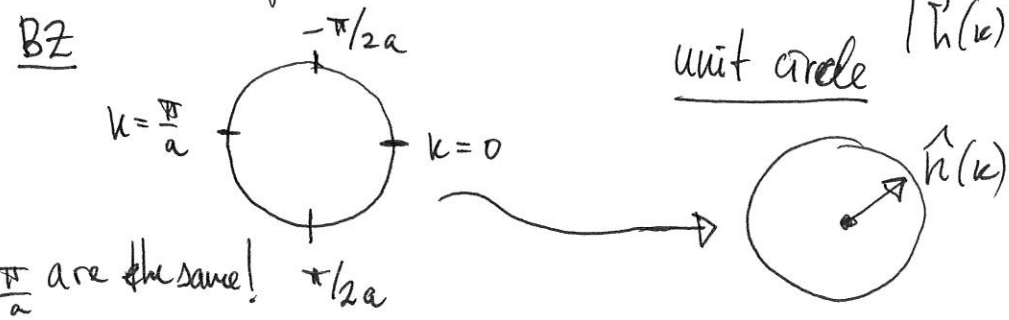
$$H = \sum_k c^+_k \left[\vec{h}(k) \cdot \vec{\sigma}_\perp\right] c_k$$

$\vec{\sigma}_\perp = \sigma^x \hat{x} + \sigma^y \hat{y}$ acts on spin degrees of freedom

$\vec{h}(k)$ = momentum-dependent (in general) Zeeman field $= h_x \hat{x} + h_y \hat{y}$

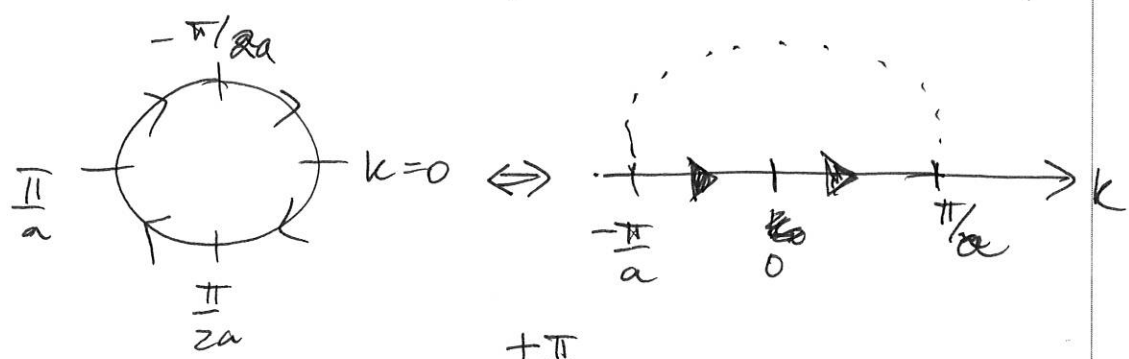
Let $\vec{h}(k) \neq 0$ for any $k \in BZ$.

Then can define unit vector $\hat{h}(k) = \dfrac{\vec{h}(k)}{|\vec{h}(k)|}$



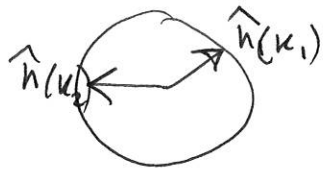
Note: $k = \frac{\pi}{a}$ and $k = -\frac{\pi}{a}$ are the same!

Space of $\vec{k}$ is a circle because for any function $f(k=\frac{\pi}{a}) = f(k=-\frac{\pi}{a})$ due to lattice's periodicity.



i.e. the ends of the BZ ($\pm\frac{\pi}{a}$) are <u>identified</u>.

Space of $\hat{n}(k)$ = unit circle



Since, by assumption, $|\hat{n}(k)| = 1$ for all k, we can ask the following question: as we sweep k from $-\frac{\pi}{a}$ to $+\frac{\pi}{a}$, how many times does $\hat{n}(k)$ wind around the unit circle? This number defines <u>topological winding number</u>.

① let $\vec{h}(k) = \varepsilon_1 \hat{x}$

$$H_1 = \sum_k c^\dagger_k \left(\varepsilon_1 \hat{x} \cdot \sigma^x \hat{x}\right) c_k =$$

$$= \sum_k \varepsilon_1 \left(c^\dagger_{k\uparrow}, c^\dagger_{k\downarrow}\right) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} c_{k\uparrow} \\ c_{k\downarrow} \end{pmatrix} =$$

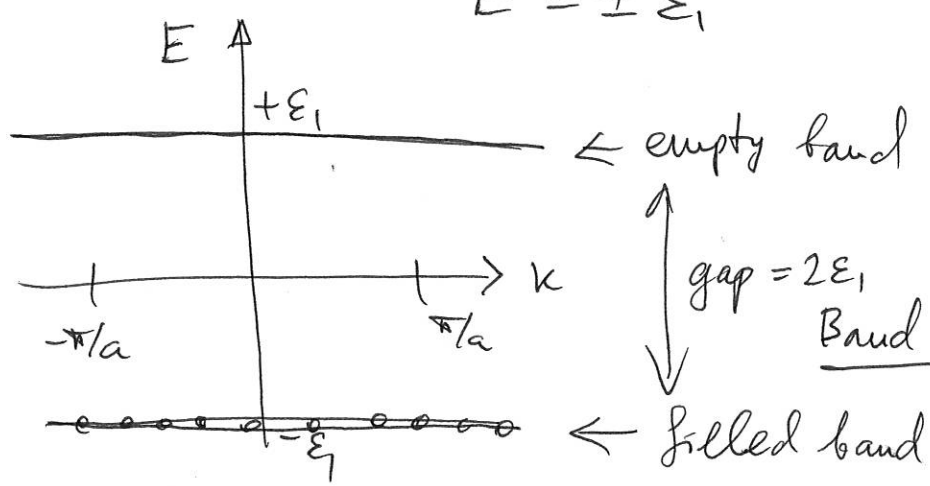
$$= \varepsilon_1 \sum_k \left(c^\dagger_{k\uparrow}, c^\dagger_{k\downarrow}\right) \begin{pmatrix} c_{k\downarrow} \\ c_{k\uparrow} \end{pmatrix} =$$

$$= \varepsilon_1 \sum_k \left(c^\dagger_{k\uparrow} c_{k\downarrow} + c^\dagger_{k\downarrow} c_{k\uparrow}\right).$$

Band structure:

$$\det \begin{pmatrix} -E & \varepsilon_1 \\ \varepsilon_1 & -E \end{pmatrix} = E^2 - \varepsilon_1^2 = 0$$

$$E = \pm \varepsilon_1$$



$\hat{n}(k) = \hat{x}$

clearly, winding number $= 0$.
"topologically trivial" phase

(2) $\vec{h}(k) = \varepsilon_2 \left(\cos(ka)\,\hat{x} + \sin(ka)\,\hat{y} \right)$

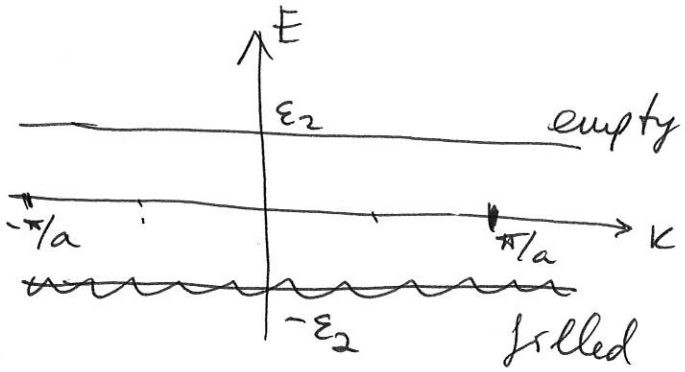
$$H_2 = \varepsilon_2 \sum_{k \in BZ} \left(c^\dagger_{k\uparrow}, c^\dagger_{k\downarrow} \right) \begin{pmatrix} 0 & \cos(ka) - i\sin(ka) \\ \cos(ka) + i\sin(ka) & 0 \end{pmatrix} \begin{pmatrix} c_{k\uparrow} \\ c_{k\downarrow} \end{pmatrix}$$

$$= \varepsilon_2 \sum_{k \in BZ} \left(c^\dagger_{k\uparrow}, c^\dagger_{k\downarrow} \right) \begin{pmatrix} 0 & e^{-ika} \\ e^{ika} & 0 \end{pmatrix} \begin{pmatrix} c_{k\uparrow} \\ c_{k\downarrow} \end{pmatrix}$$

$$\det \begin{pmatrix} -E & \varepsilon_2 e^{-ika} \\ \varepsilon_2 e^{ika} & -E \end{pmatrix} = E^2 - \varepsilon_2^2 e^{-ika} \cdot e^{ika} = 0$$

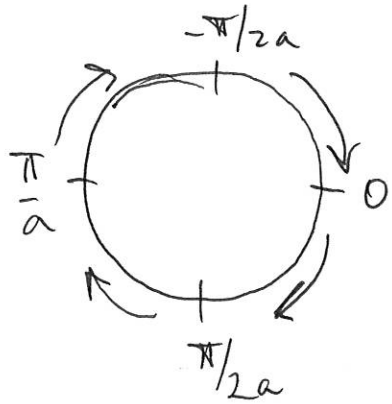
$$E = \pm \varepsilon_2 .$$

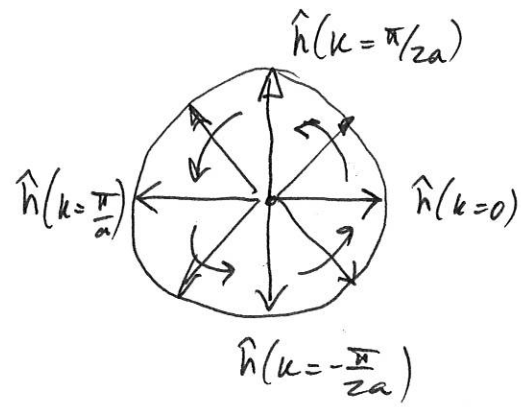
identical with H_1 ~~spectrum~~ band structure.



$2\varepsilon_2$ Band insulator

But $\hat{n}(k) = \cos(ka)\,\hat{x} + \sin(ka)\,\hat{y}$





Electron's spin winds as it moves across the BZ!

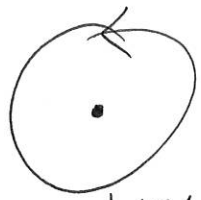
winding number $= 1$.

"topologically non-trivial" phase

Simple <u>Mathematics</u> of winding number:

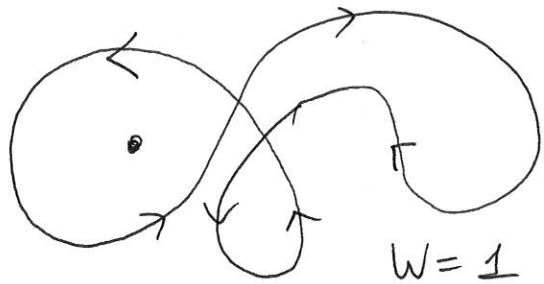
$W=0$

winding number of a closed curve on a plane around a given point is ~~given by~~ an <u>integer</u> equal to total number of times the curve travels (counterclockwise) around the point.



$W=1$

$W=-1$



$W=1$

Once $W \neq 0$, the curve cannot be deformed smoothly so as to removed from that point
$\rightarrow$ hence, W represents topological invariant.

Let the curve $\vec{r}(t) = (x(t), y(t))$ be defined parametrically. (think of t as a time, for example.)

$$\vec{r}(t=0) = \vec{r}(t=1) \quad (\text{the curve is } \underline{\text{closed}})$$

In polar coordinates, $\vec{r}(t) \mapsto \begin{cases} r(t) \equiv |\vec{r}(t)| > 0 \\ \theta(t) \end{cases}$

So that $\vec{r}(t) = r(t)\, e^{i\theta(t)}$

$$\begin{cases} \mathrm{Re}\left(\vec{r}(t)\right) = x(t) \\ \mathrm{Im}\left(\vec{r}(t)\right) = y(t) \end{cases}$$

Since $\vec{r}(t=0) = \vec{r}(t=1)$, $\theta(t=0)$ and $\theta(t=1)$ may differ only by (Integer $\cdot\, 2\pi$)

This integer is the winding number : $W = \dfrac{\theta(1)-\theta(0)}{2\pi}$

<u>Integral representation</u>

$$\left.\begin{array}{l} x = r\cos\theta \\ y = r\sin\theta \end{array}\right\} \quad \tan\theta = \frac{y}{x} \quad , \quad \theta = \arctan\left(\frac{y}{x}\right)$$

$$\theta(t) = \arctan\left(\frac{y(t)}{x(t)}\right)$$

$$\frac{d\theta(t)}{dt} = \frac{1}{1+\left(\frac{y(t)}{x(t)}\right)^2} \frac{d}{dt}\left(\frac{y(t)}{x(t)}\right) =$$

$$= \frac{x^2(t)}{x^2(t)+y^2(t)} \left(\frac{1}{x(t)}\frac{dy(t)}{dt} - \frac{y(t)}{x^2(t)}\frac{dx(t)}{dt}\right) =$$

$$= \frac{1}{x^2(t)+y^2(t)} \left(x(t)\frac{dy(t)}{dt} - y(t)\frac{dx(t)}{dt}\right)$$

$$\Rightarrow \quad d\theta = \frac{1}{r^2(t)}\left(x(t)\,dy(t) - y(t)\,dx(t)\right) dt$$

$$\Rightarrow \quad \theta(t=1) - \theta(t=0) = \oint_C \left(\frac{x}{r^2}\,dy - \frac{y}{r^2}\,dx\right)$$

$$W = \frac{1}{2\pi} \int_0^1 dt \left(\frac{x}{r^2} \frac{dy}{dt} - \frac{y}{r^2} \frac{dx}{dt} \right)$$

Note that $x \frac{dy}{dt} - y \frac{dx}{dt} = \hat{z} \cdot \vec{r} \times \partial_t \vec{r}$

Indeed: $\hat{z} \cdot \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ x(t) & y(t) & 0 \\ \frac{dx}{dt} & \frac{dy}{dt} & 0 \end{vmatrix} = x \frac{dy}{dt} - y \frac{dx}{dt}$

$$W = \frac{1}{2\pi} \int_0^1 dt \, \hat{z} \cdot \vec{r}(t) \times \partial_t \vec{r}(t)$$

Apply to our problem: $\begin{cases} t \to k \\ t \in (0,1) \; ; \; k \in BZ = (-\frac{\pi}{a}, \frac{\pi}{a}). \\ \vec{r}(t) \to \hat{n}(k) \end{cases}$

Since $\hat{n}(k)$ is unit vector, $\vec{r}^{\,2}(t) = x^2 + y^2 \longrightarrow \hat{n}^2 = 1$.

Then $W = \frac{1}{2\pi} \int_{-\pi/a}^{\pi/a} dk \; \hat{z} \cdot \hat{n}(k) \times \frac{d\hat{n}}{dk} =$

$$= \frac{1}{2\pi} \int_{-\pi/a}^{\pi/a} dk \left(n_x \frac{dn_y}{dk} - n_y \frac{dn_x}{dk} \right)$$

For $\hat{n} = \hat{n}_1(k) = \hat{x}$ $\quad W = 0$

For $\hat{n} = \hat{n}_2(k) = \cos(ka)\hat{x} + \sin(ka)\hat{y}$,

$$W = \frac{1}{2\pi} \int_{-\pi}^{\pi} d(ka) \left(\cos(ka) \frac{d\sin(ka)}{d(ka)} - \sin(ka) \frac{d\cos(ka)}{d(ka)} \right) = \frac{1}{2\pi} \int_{-\pi}^{\pi} dk \left(\cos^2 k + \sin^2 k \right) = +1.$$